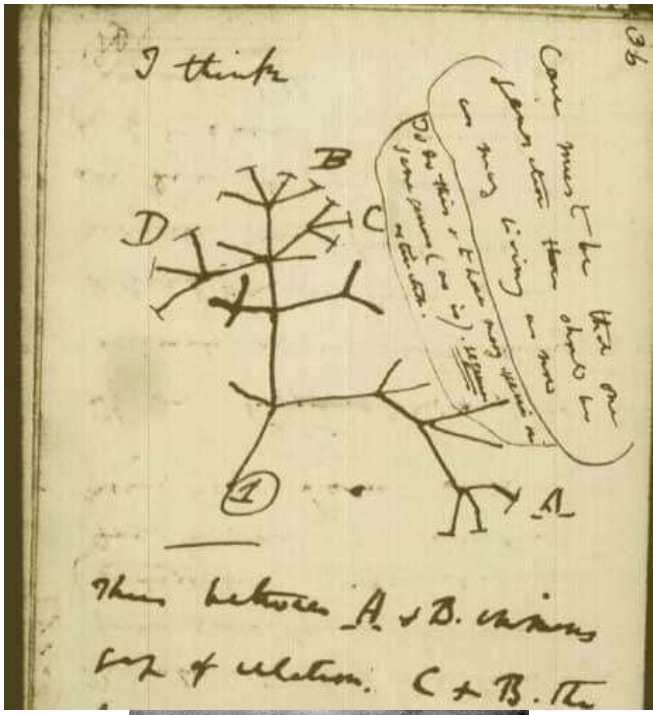
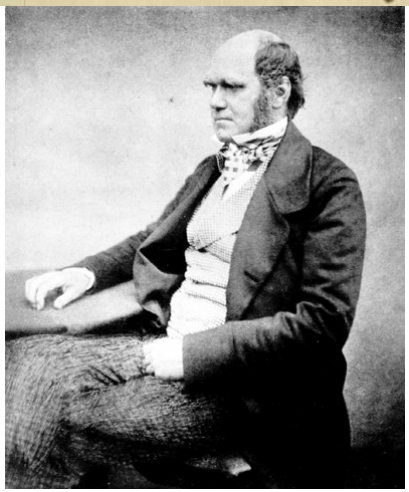
Evolutionary Theory: Evidence

Charles Darwin (1809-1882)

- Father of the theory of Evolution
- Suggested that **natural selection** is the mechanism by which species evolve over geologic time.
- Proposed **Descent with Modification**:
  - *All organisms on Earth are related through some unknown ancestral type that lived long ago.*

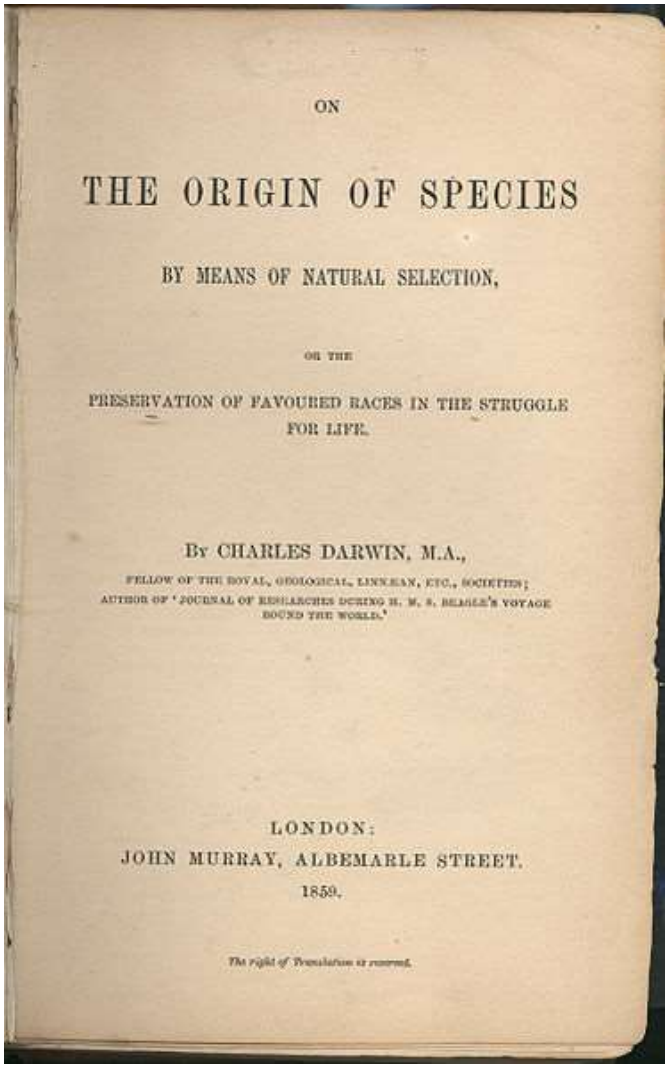




The Origin of Species (1859)

Darwin developed two main ideas:

- Evolution explains life's **unity and diversity**
- **Natural selection** is a cause of adaptive evolution



ON

THE ORIGIN OF SPECIES

BY MEANS OF NATURAL SELECTION,

OR THE

PRESERVATION OF FAVOURED RACES IN THE STRUGGLE FOR LIFE.

BY CHARLES DARWIN, M.A.,

FELLOW OF THE ROYAL, GEOLOGICAL, LINNÆAN, ETC., SOCIETIES;
AUTHOR OF 'JOURNAL OF RESEARCHES DURING H. M. S. BEAGLE'S VOYAGE ROUND THE WORLD.'

LONDON:
JOHN MURRAY, ALBEMARLE STREET.
1859.

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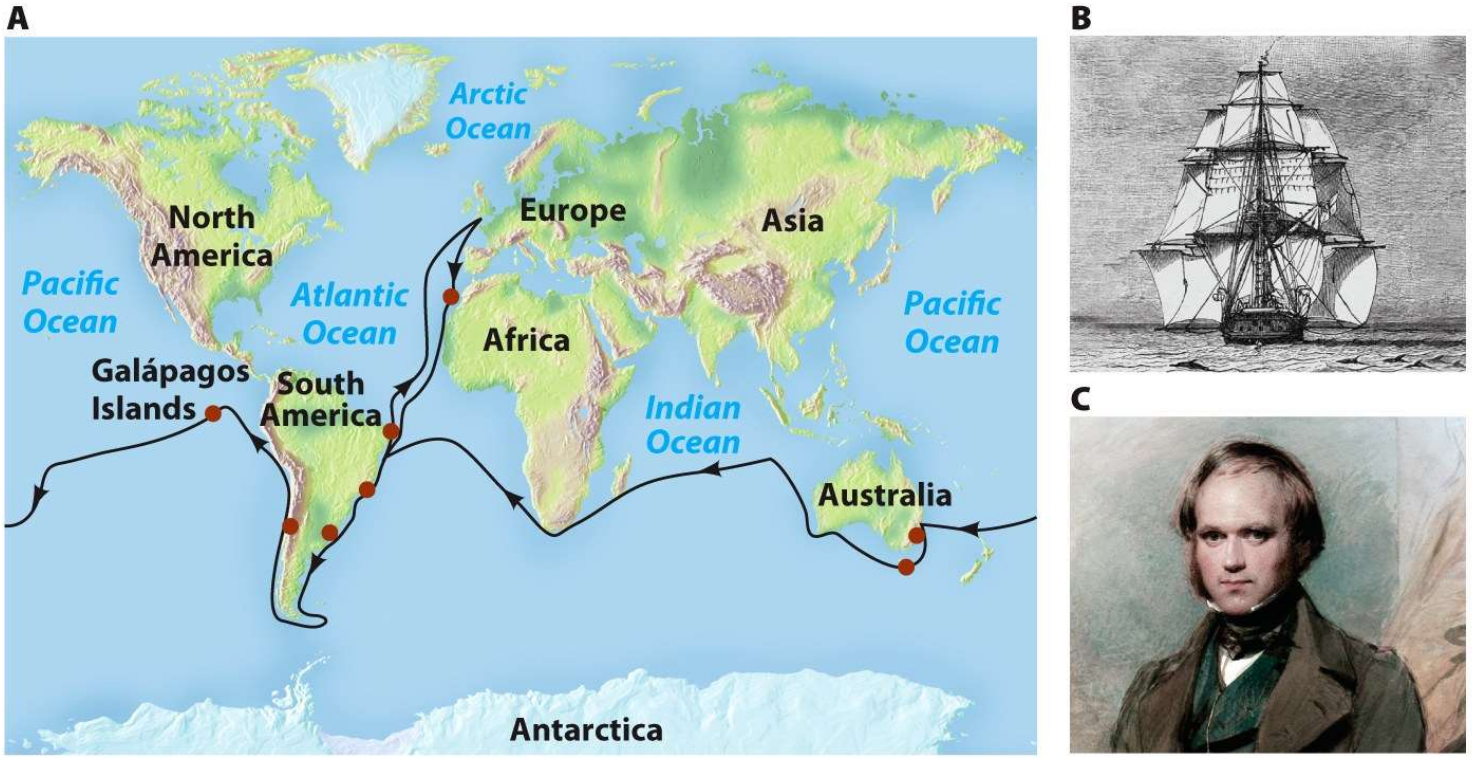
Darwin – Voyage of the Beagle 1831-1836



Evolution, 1/e Figure 2.12
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Origin of the Species
-Charles Darwin, 1859

Full Title:

On the Origin of the Species by Means of Natural Selection, or the Preservation of the Favoured Races in the Struggle for Life

Alfred Wallace

-- was born in Wales in 1823. He has been described variously as a naturalist, a geographer, and a social critic.

-- traveled to Brazil and various islands of the Malay Archipelago that make up modern-day Indonesia and the Philippines, where he collected thousands of specimens of insects, birds, and other animals.

-- his observations led him to the conclusion that living things change over long periods of time—**they evolve.**

His research on the geographic distribution of animals provided critical support for his evolutionary theories and led him to draw a boundary line through Southeast Asia that divides Asian and Australian animal groups. Wallace's Line, as it was later named, runs from the Indian Ocean to the Philippine Sea. It signifies the unexpected distribution of animals on either side of the line.

What is the evidence for evolution?

1) Fossil Record

Evidence that species change over time.

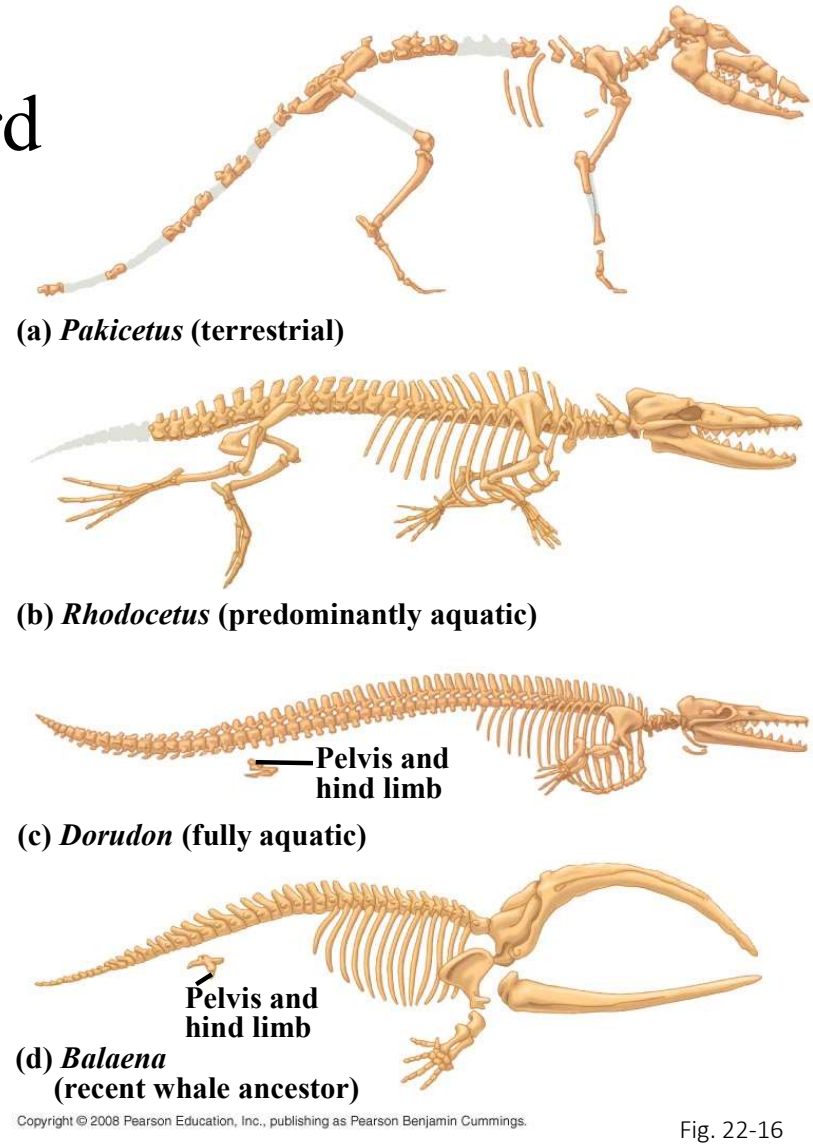

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Fig. 22-16

Fossil Record
a) **Extinction** has occurred. Evidence of <u>change over time in species composition</u>.

Figure 2-8 Evolutionary Analysis, 4/e
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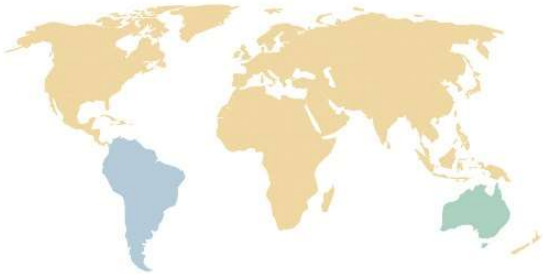
Fossil Record

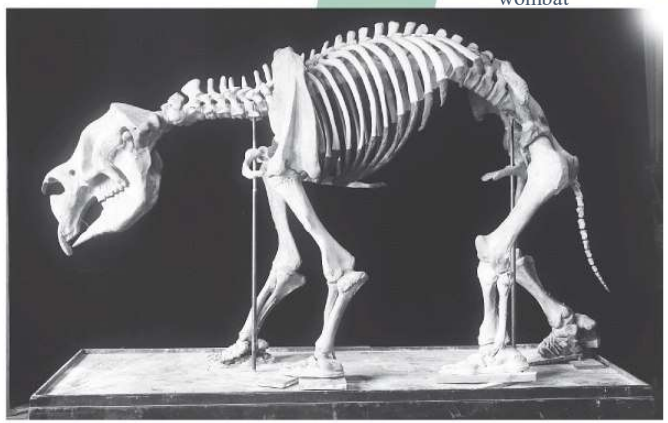
b) Fossil and extant species of the same region resemble one another (**law of succession**).

pygmy armadillo



wombat

glyptodont



Diprotodon

Figure 2-9 Evolutionary Analysis, 4/e

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Extinct species are related to extant species by descent

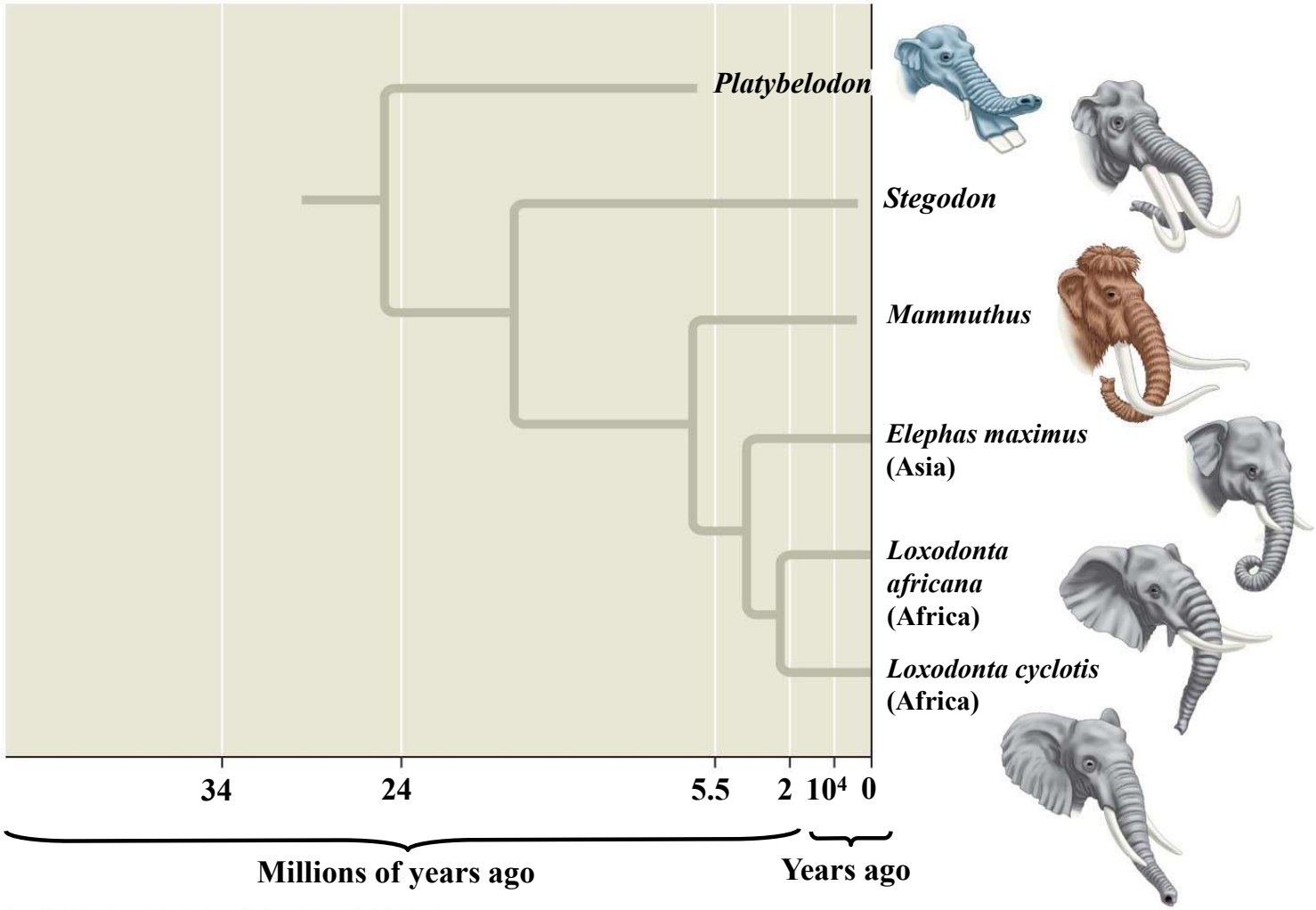

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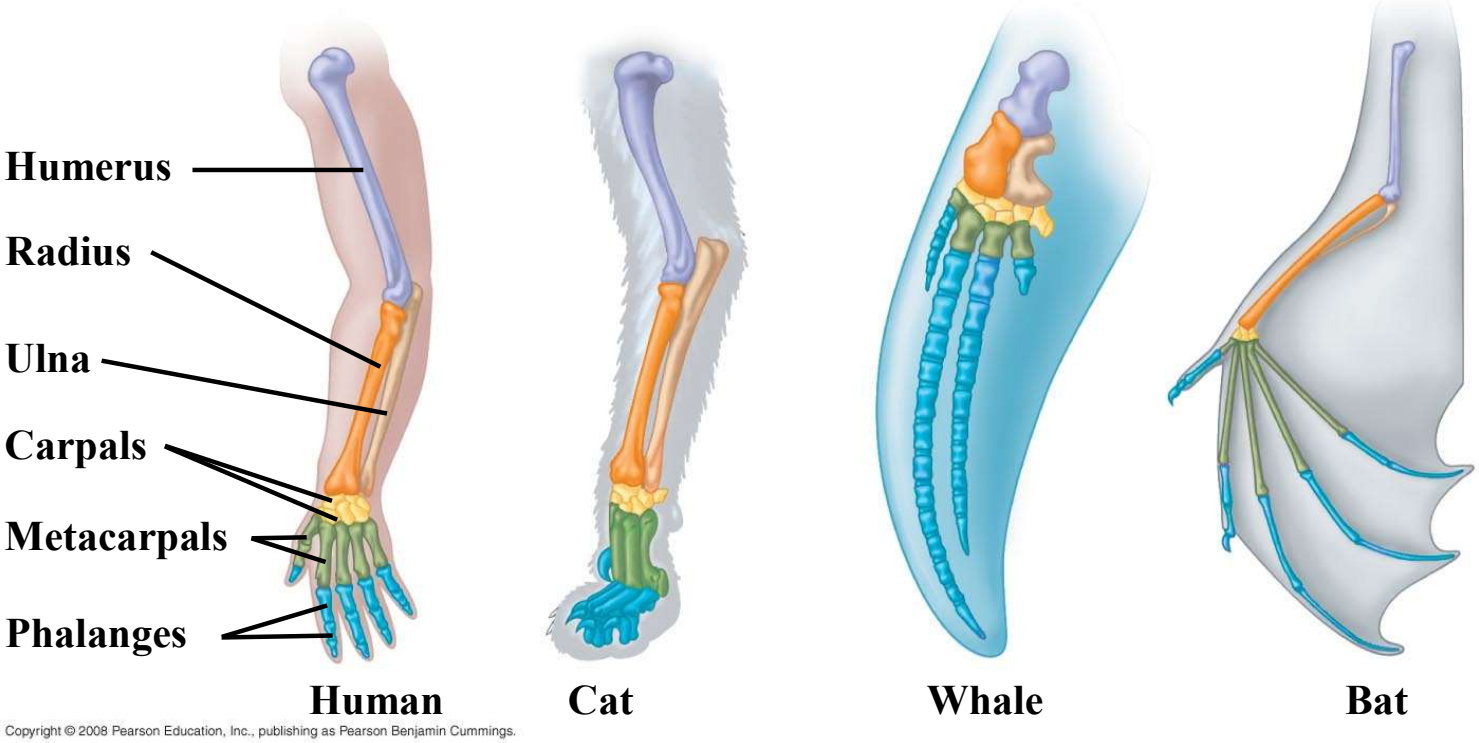
Fig. 22-8a

EVIDENCE:
2) Comparative anatomy / morphology

<u>Homology</u> = similarity due to common ancestry!
Mature anatomy



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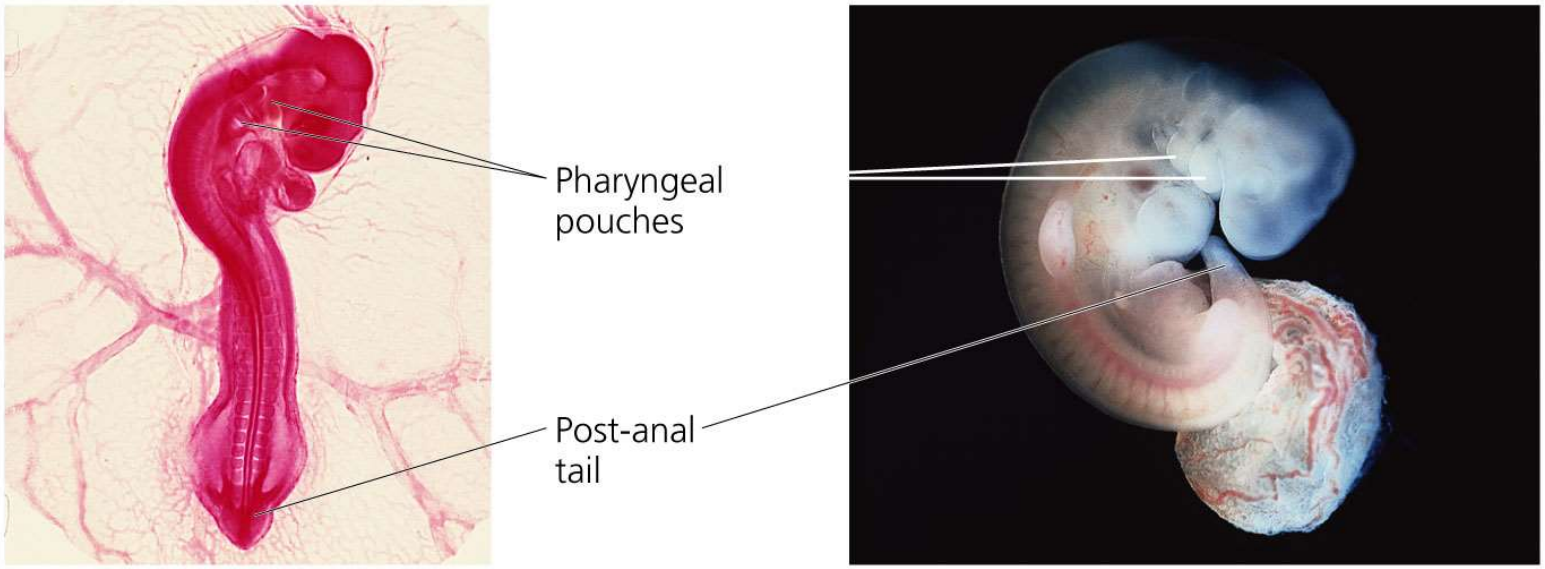
EVIDENCE:
3) Comparative development

Developmental similarity is due to common ancestry (homology).



Chick embryo (LM) Human embryo

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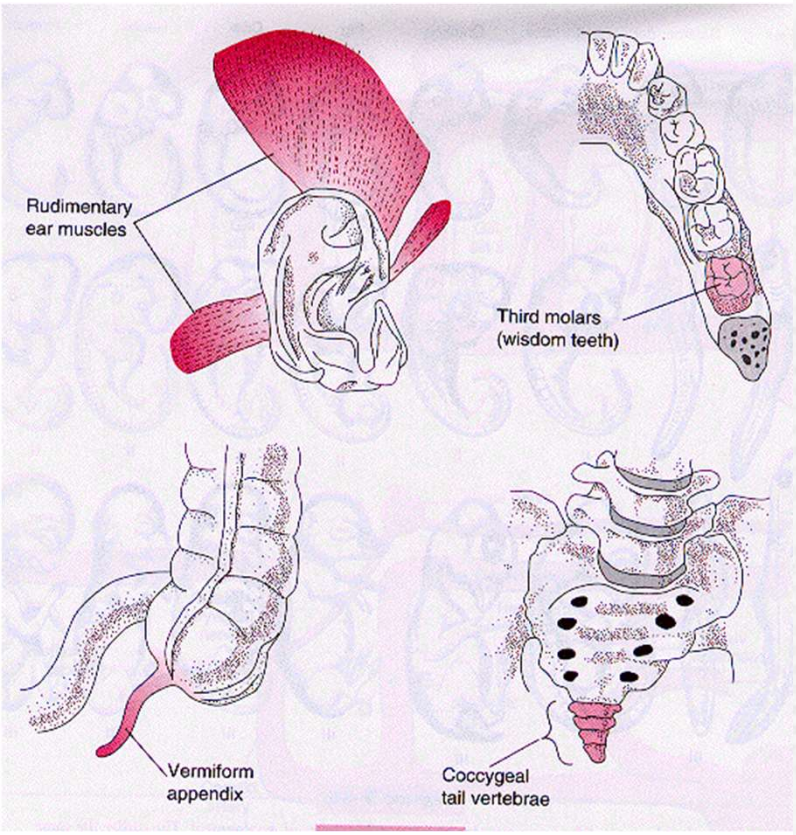
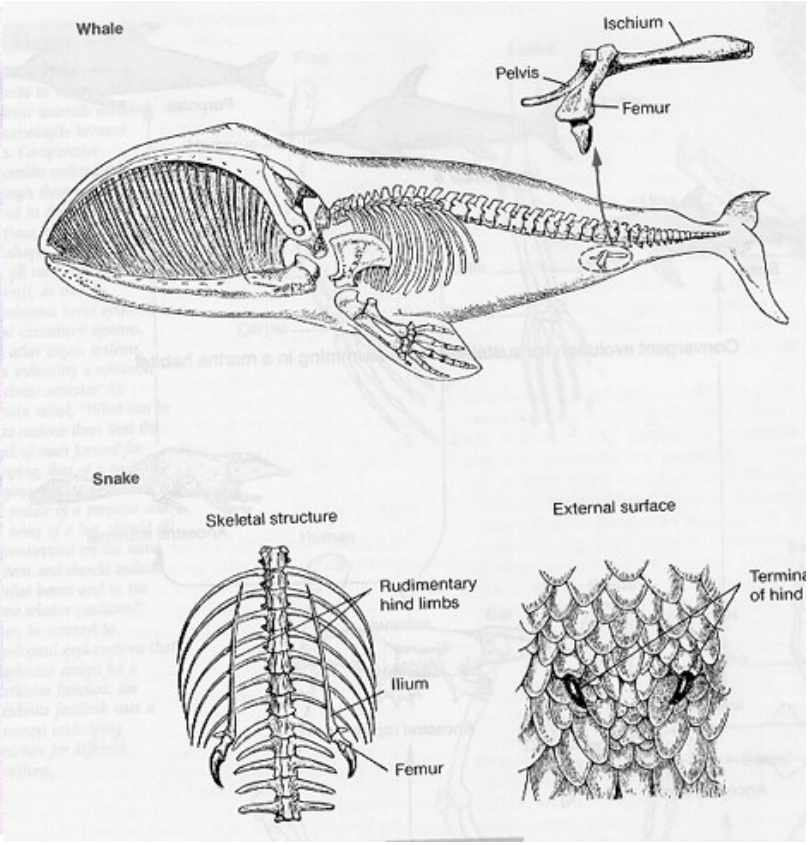
EVIDENCE:

4) Vestigial organs

Vestigial organs are remnants of common ancestry.

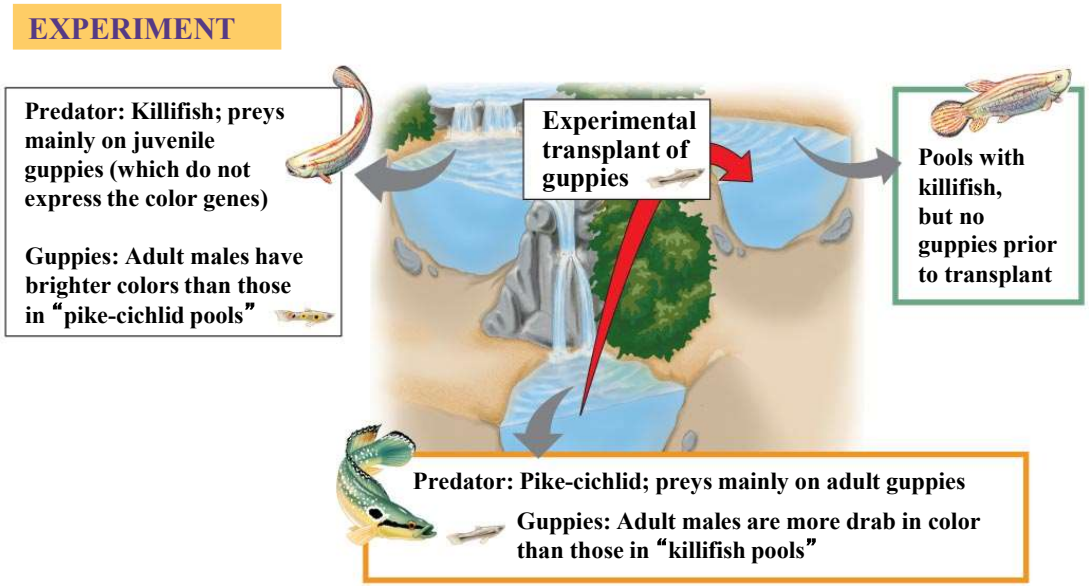




EVIDENCE:

5) Direct observation of evolutionary change.

Experiments by John Endler on guppies



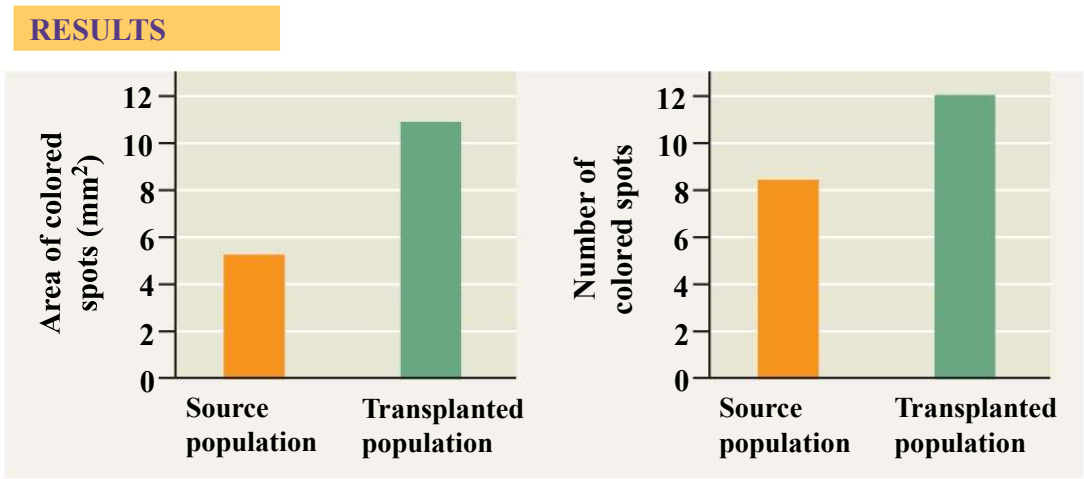


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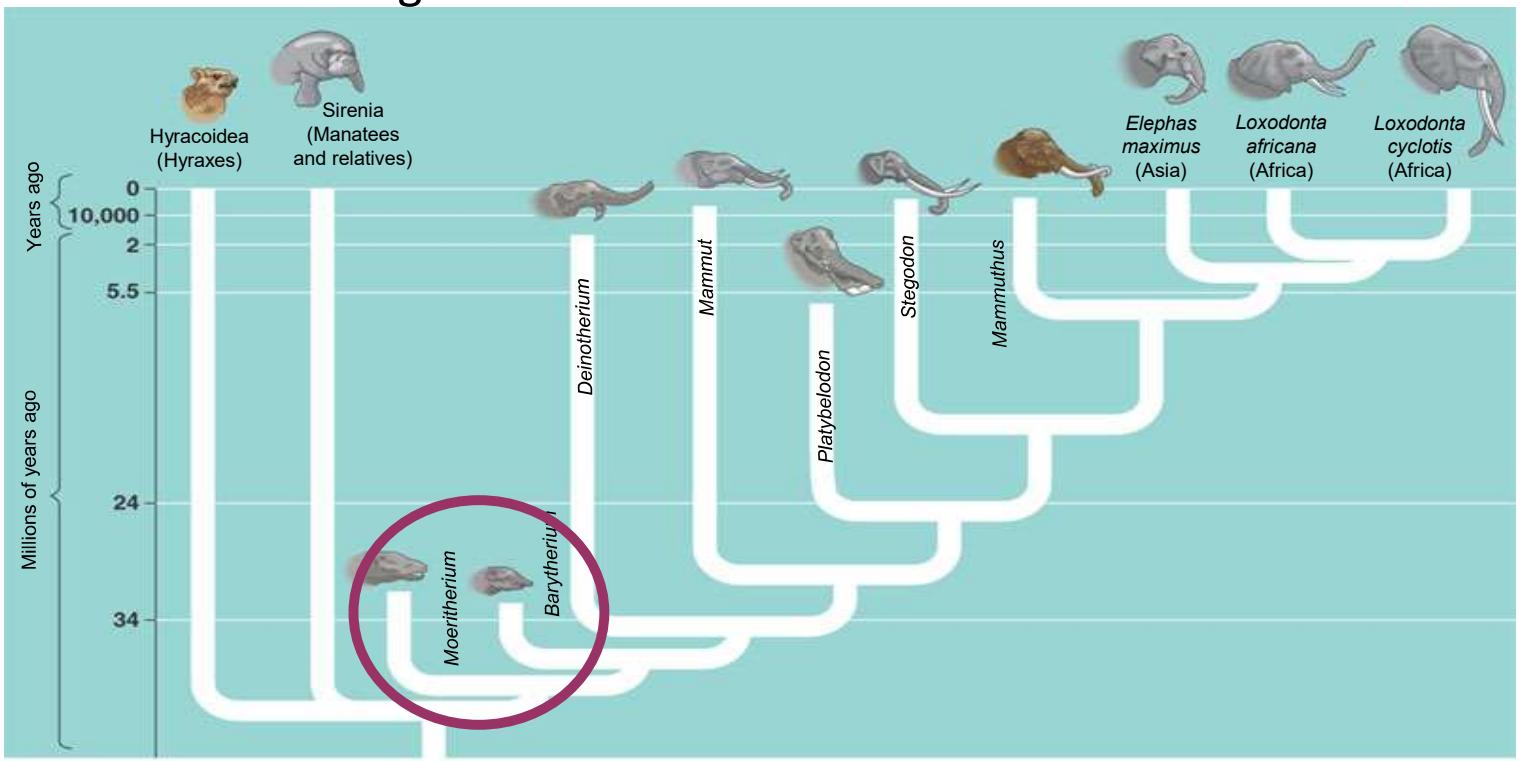
Fig. 22-13

To Darwin, the history of life is like a tree.

→multiple branchings from a common trunk to the tips of the youngest twigs that represent the diversity of living organisms



The Six Main Points of Darwin's Theory of Evolution

Observations and Inferences

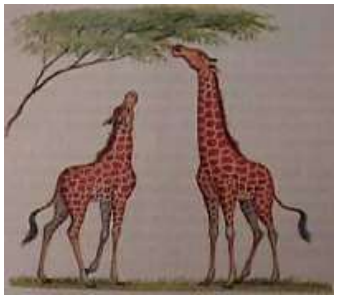
Natural selection does not grant organisms what they "need".

1. Overproduction*

- Most species produce far more offspring than are needed to maintain the population.

- Species populations remain more or less constant (“stable”) because a <u>small fraction</u> of offspring live long enough to reproduce.

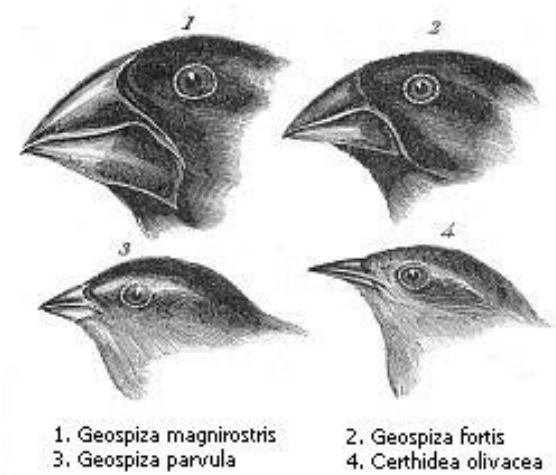
2. Competition*



- Living space and food are limited, so offspring from each generation must compete among themselves in order to live.

- Only a small fraction can possibly survive long enough to reproduce.

3. Genetic Variation*



Finches from Galapagos Archipelago

- Characteristics in individuals in any species are not exactly alike.
  - Ex: Differences for *Homo sapiens* (humans) -can not be exact in size or shape of body, strength in running, or resistance to disease.
- These differences are considered to be the variations ***within*** a species. *What causes slight variations between individuals?*

4. Adaptation

An adaptation: any **inherited trait** that **increases** an organisms' chance of survival and reproduction in a given environment.

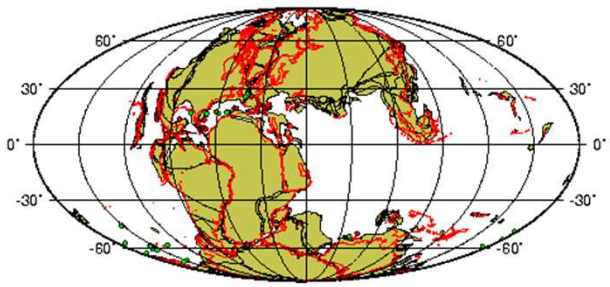
*The word "adapt" is a **VERBAL SHORTCUT***

5. Natural Selection*

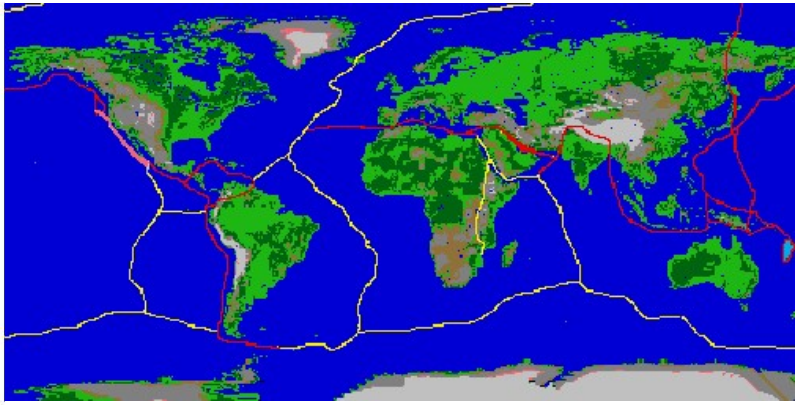
- **Nature**/environment **selects** for living organisms with better suited **inherited** traits to survive and reproduce.
- Variation **caused by** _____within a species (ex: giraffe) *may* make them better "equipped" for survival.
- Offspring inherit these better traits, and **<u>as a whole</u>** the population improves *for that particular environment*.

5. Natural Selection, cont.

- Natural Selection does not move in a pre-determined direction! The changing earth determines what will and can survive.



150 My Reconstruction



6. Speciation

- Over many generations, favorable adaptations (in a *particular* environment) gradually accumulate in a species and “bad” ones (in a *particular* environment) disappear.
- Eventually, accumulated changes become so great, the result is a **new species**.
- Formation of a new species is called “Speciation” and it takes *many, many* generations to do.

The four factors*

1. Overproduction
2. Competition
3. Genetic Variation
4. Natural Selection

Biological Evolution is a consequence of these 4 factors - they work together to impact any living population

Additional evidence supporting evolutionary theory

1. today the layers of rock can be accurately dated by strata and by radioactive decay methods we can see that species have been altered over geologic time the fossil record shows clearly that all organisms did not appear at the same time. Many that once existed have become extinct

~99% of all life that ever existed on earth is now extinct

the fossil record also shows there has been an orderly sequence of species change and replacement over billions of years from the simplest forms of live to the most complex:

3.5 BY bacteria
1.5 BY eucaryotes
500 MY animals
400 MY plants
190 MY 1st flowering plants
65 MY modern plants, primates
15 MY hominids

2. also, we have found fossils of many “intermediate forms” between major groups in the fossil record:

eg. between fish and amphibians
eg. between dinosaurs and birds
eg. evolutionary stages of the horse, elephant, etc
eg. human ancestors

in a few cases we have essentially every major step in the evolutionary process from one species to another

eg. 1 snail species split into 2 in South American Lake sediment (year by year evolutionary record through fossils)

3. We have learned the science of genetics and can explain how mutations occur and how they are passed on (this process was completely unknown to Darwin)

4. In modern times we have added a massive amount of **molecular evidence** that supports evolutionary theory.
Similarities and differences in biochemistry correlates with assumed evolutionary relationships
The more closely related an animal is the more similar its biochemistry:

eg. DNA
eg antibodies
eg. protein structure

eg. nematode worm shares 40% of its DNA with us
eg. chimpanzees and humans share 98% of their DNA
eg. all humans share 99% of our DNA
eg. closer relatives 99.5%

We can even quantify the degree of difference and the evolutionary timelines for virtually all forms of life virtually everything we know about the natural world

biology
geology
chemistry
physics
astronomy
contribute to our current understanding of the process of evolution